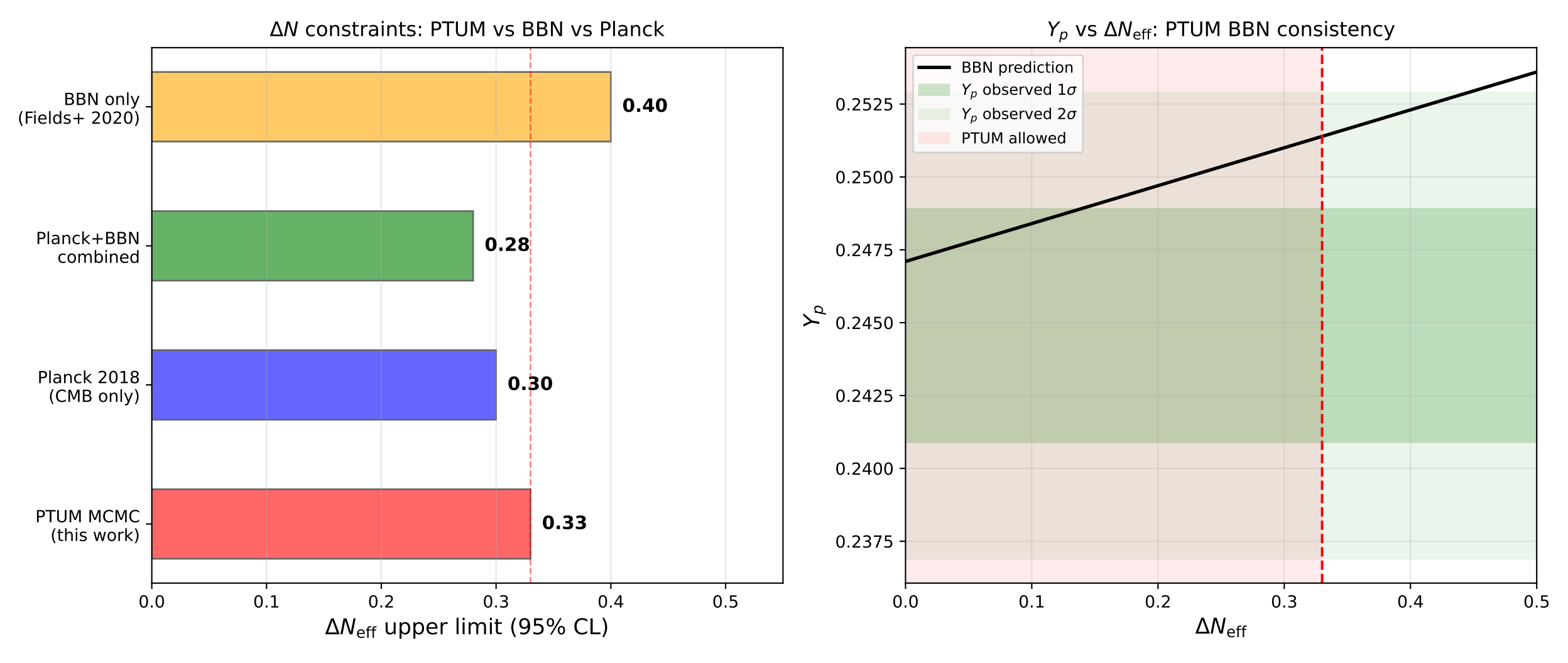
ΔN constraints: PTUM vs BBN vs Planck
BBN only (Fields+ 2020)
Planck+BBN combined
Planck 2018 (CMB only)
PTUM MCMC (this work)
0.40
0.28
0.30
0.33
0.0
0.1
0.2
0.3
0.4
0.5
ΔN_eff upper limit (95% CL)
Y_p vs ΔN_eff: PTUM BBN consistency
BBN prediction
Y_p observed 1σ
Y_p observed 2σ
PTUM allowed
0.2525
0.2500
0.2475
0.2450
0.2425
0.2400
0.2375
Y_p
0.0
0.1
0.2
0.3
0.4
0.5
ΔN_eff